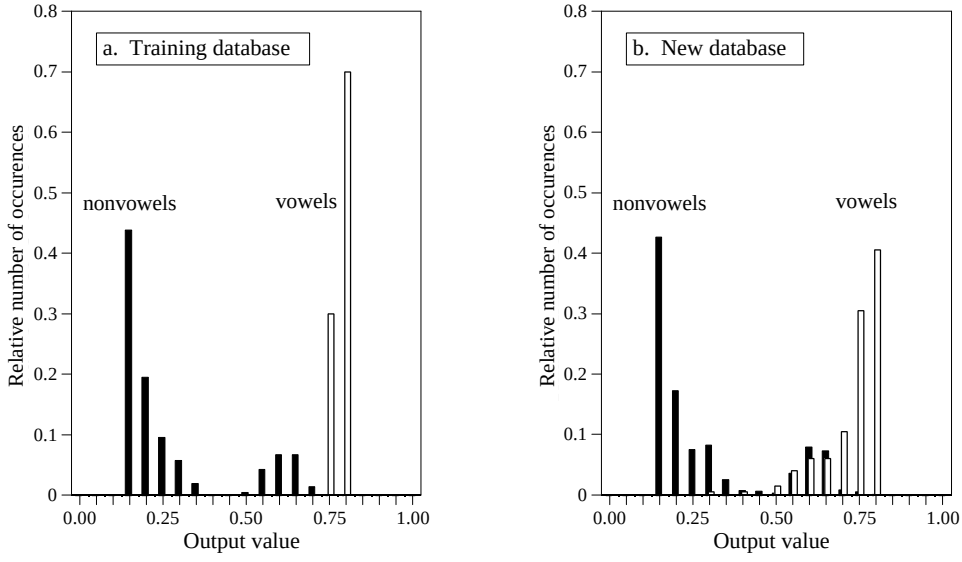

FIGURE 26-11
Neural network performance. These are histograms of the neural network's output values, (a) for the training data, and (b) for the remaining images. The neural network performs better with the training data because it has already seen the answers to the test.

other two. This is a matter of random chance depending on the initial weights used. At one threshold setting, the neural network designed in trial "b" can detect 24 out of 25 targets (i.e., 96% of the vowel images), with a false alarm rate of only 1 in 25 nontargets (i.e., 4% of the nonvowel images). Not bad considering the abstract nature of this problem, and the very general solution applied.

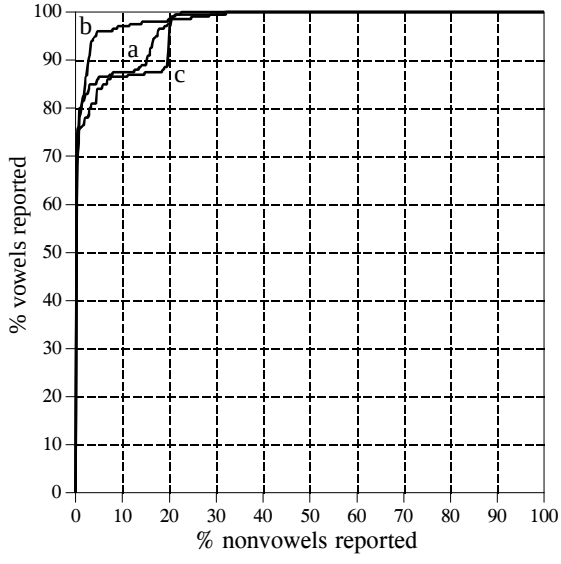

FIGURE 26-12
ROC analysis of neural network examples. These curves compare three neural networks designed to detect images of vowels. Trial (b) is the best solution, shown by its curve being closer to the upper-left corner of the graph. This network can correctly detect 24 out of 25 targets, while providing only 1 false alarm for each 25 nontargets. That is, there is a point on the ROC curve at $x = 4\%$ and $y = 96\%$